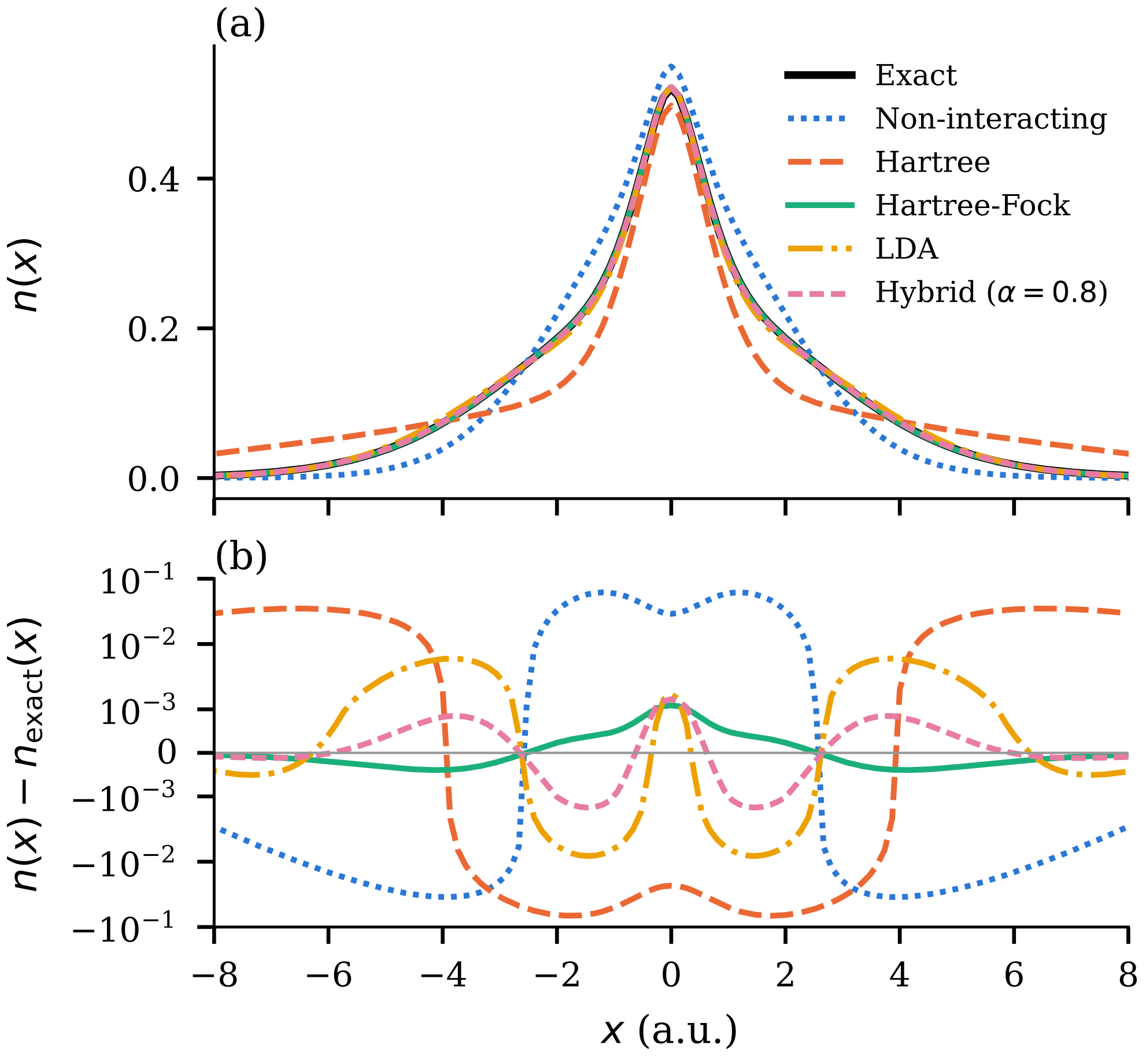
(a)
Exact
Non-interacting
Hartree
Hartree-Fock
LDA
Hybrid ($\alpha = 0.8$)
$n(x)$
0.0
0.2
0.4
(b)
$n(x) - n_{\text{exact}}(x)$
10^{-1}
10^{-2}
10^{-3}
0
-10^{-3}
-10^{-2}
-10^{-1}
−8
−6
−4
−2
0
2
4
6
8
x (a.u.)